Advanced Bank Loan Data Analysis: Empowering Decision-Making through Dynamic Insights

Problem Statement: Advanced Dashboard Development for Loan Data Analysis. The current approach to loan data analysis lacks depth and interactivity, hindering the ability to derive meaningful insights for informed decision-making. Traditional reporting methods fall short in providing a comprehensive view of lending operations, borrower behavior, and loan performance metrics. There is a pressing need for advanced dashboard design to address these limitations and unlock the full potential of our loan data.

Objective: Our objective is to craft a suite of interconnected dashboards that deliver dynamic and comprehensive insights into our loan data, empowering decision-makers with actionable intelligence derived from robust data analysis. These dashboards are meticulously designed to offer a holistic perspective on our lending operations, borrower demographics, loan performance, and financial metrics, facilitating strategic decision-making based on data-driven insights.

Dashboard 1: Executive Summary

Purpose: This dashboard provides an overview of critical Key Performance Indicators (KPIs) essential for evaluating the overall efficiency and performance of our lending endeavors.

Key Features:

- **Total Loan Application**s: Monitor total and Month-to-Date (MTD) applications, scrutinizing Month-over-Month (MoM) trends for actionable insights.
- **Total Funded Amount**: Evaluate the cumulative disbursed funds and track MTD disbursements, analyzing MoM variations to discern funding patterns.
- **Total Amount Received**: Assess cash inflows via total and MTD received amounts from borrowers, scrutinizing MoM fluctuations for financial health assessment.
- **Average Interest Rate**: Determine the cost of lending by computing and tracking the average interest rate across all loans, including MTD and MoM changes.
- **Average Debt-to-Income Ratio (DTI)**: Evaluate borrowers' financial resilience by calculating the average DTI for all loans and monitoring MTD and MoM changes.

BANK LOAN REPORT | SUMMARY

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Good vs Bad Loan
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Total loan Applications	Total Funded Amount	Total Amount Received	Average Interest Rate	Average DTI
38.6K	$435.8M	$435.8M	12.0%	13.3%
MTD 0.4M%	MTD $54.0M	MTD $58.1M	MTD 12.4%	MTD 13.7%
MoM 6.9%	MoM 13.0%	MoM 15.8%	MoM 3.5%	MoM 3%

Good Loan Issued

86.2%

good_loan_appl
33.2K

good_loan_funded_amt
$370.2M

good_loan_received_amt
$435.8M

Good Loan Issued

13.8%

bad_loan_appl
5.3K

bad_loan_funded_amt
$65.5M

bad_loan_received_amt
$37.284763M

Loan status	Total loan Appl	Total Funded Amt	Total Received Amt	MTD_funded_amt	MTD_received_amt	Avg Interest Rate	Avg DTI
Charged Off	5333	$65,532,225	$37,284,763	$8,732,775	$5,324,211	13.88%	14.00%
Current	1098	$18,866,500	$24,199,914	$3,946,625	$4,934,318	15.10%	14.72%
Fully Paid	32145	$351,358,350	$411,586,256	$41,302,025	$47,815,851	11.64%	13.17%
Total	38576	$435,757,075	$473,070,933	$53,981,425	$58,074,380	12.05%	13.33%

BANK LOAN REPORT | OVERVIEW

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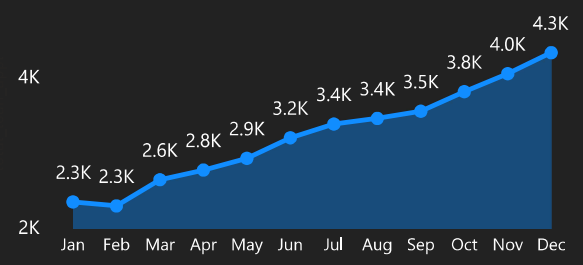
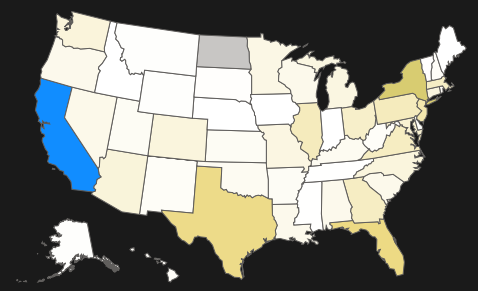
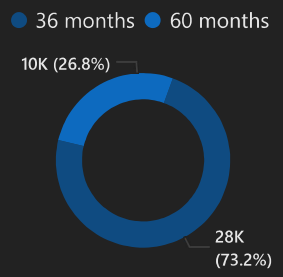
Total loan appl by Month



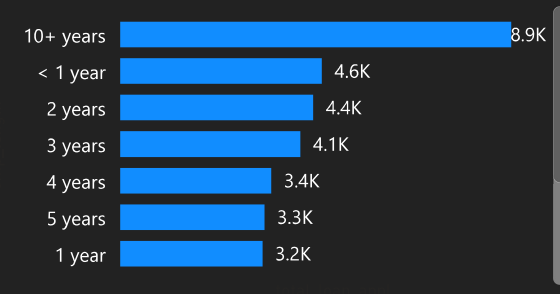
Total loan Application by State



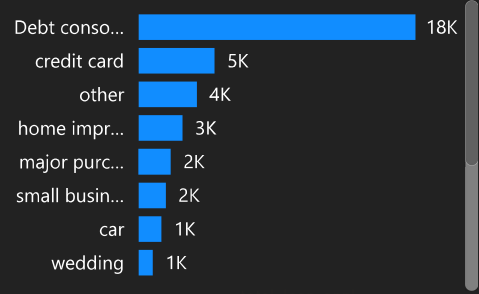
total_loan_appl by Term



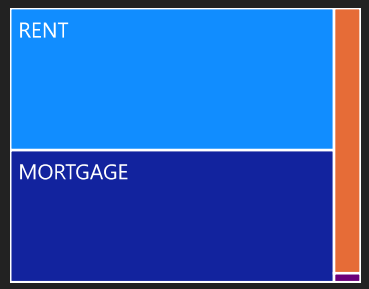
Total Loan Application by Emp length



Total loan appl by purpose



Total loan appl by home ownership



BANK LOAN REPORT | DETAILS

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Total loan Applications		Total Funded Amount		Total Amount Received		Average Interest Rate		Average DTI	
38.6K		$435.8M		$435.8M		12.0%		13.3%	
MTD	MoM	MTD	MoM	MTD	MoM	MTD	MoM	MTD	MoM
0.4M%	6.9%	$54.0M	13.0%	$58.1M	15.8%	12.4%	3.5%	13.7%	3%

ID	Purpose	Home ownership	Grade	Sub_grade	Issue Date	Funded Amt	Avg Interest Rate	Installment	Total Rece
989285	Debt consolidation	RENT	G	G1	10/11/2021	$35,000	22.74%	981.45	
812976	Debt consolidation	MORTGAGE	G	G2	8/11/2021	$35,000	22.48%	976.24	
972576	credit card	MORTGAGE	F	F5	10/11/2021	$35,000	22.35%	973.64	
874599	Debt consolidation	MORTGAGE	G	G3	9/11/2021	$35,000	22.85%	983.66	
768930	small business	MORTGAGE	F	F3	6/11/2021	$35,000	20.99%	946.68	
674448	Debt consolidation	MORTGAGE	G	G2	2/11/2021	$35,000	20.48%	936.66	
914211	Debt consolidation	MORTGAGE	F	F1	10/11/2021	$35,000	20.89%	944.71	
772157	small business	RENT	G	G1	6/11/2021	$35,000	22.11%	968.86	
1057770	Debt consolidation	MORTGAGE	E	E5	12/11/2021	$35,000	20.30%	933.14	
833224	Debt consolidation	MORTGAGE	F	F2	8/11/2021	$35,000	20.62%	939.41	
698163	home improvement	MORTGAGE	G	G2	3/11/2021	$35,000	20.48%	936.66	
762870	Debt consolidation	OWN	E	E4	5/11/2021	$35,000	19.29%	913.52	
768153	home improvement	MORTGAGE	F	F2	6/11/2021	$35,000	20.62%	939.41	
1057239	Debt consolidation	RENT	E	E3	12/11/2021	$35,000	19.42%	916.03	
1034299	credit card	MORTGAGE	E	E3	12/11/2021	$35,000	19.42%	916.03	
1008529	Debt consolidation	MORTGAGE	E	E5	11/11/2021	$35,000	20.30%	933.14	
733401	credit card	RENT	G	G4	4/11/2021	$35,000	21.22%	951.21	